

And one final iteration:

- $a = 0.375, b = 0.5$ where $f(0.375) < 0$ and $f(0.5) > 0$.
- Test the point in the middle $x = \frac{0.375+0.5}{2} = 0.4375$

$$f(0.4375) = 0.0718932843 > 0$$

- So our new interval will be $[0.375, 0.4375]$ since the function is negative at $x = 0.375$ and positive at $x = 0.4375$

So without much work we know the location of a zero inside a range of length $0.0625 = 2^{-4}$. Each iteration will halve the length of the range and we keep going until we reach the precision we need, though it is much easier to program a computer to do it.

Example 1.6.15

1.7 ▲ (Optional) — Making the Informal a Little More Formal

As we noted above, the definition of limits that we have been working with was quite informal and not mathematically rigorous. In this (optional) section we will work to understand the rigorous definition of limits.

Here is the formal definition — we will work through it all very slowly and carefully afterwards, so do not panic.

Definition 1.7.1.

Let $a \in \mathbb{R}$ and let $f(x)$ be a function defined everywhere in a neighbourhood of a , except possibly at a . We say that

the limit as x approaches a of $f(x)$ is L

or equivalently

as x approaches a , $f(x)$ approaches L

and write

$$\lim_{x \rightarrow a} f(x) = L$$

if and only if for every $\epsilon > 0$ there exists $\delta > 0$ so that

$$|f(x) - L| < \epsilon \text{ whenever } 0 < |x - a| < \delta$$

Note that an equivalent way of writing this very last statement is

$$\text{if } 0 < |x - a| < \delta \text{ then } |f(x) - L| < \epsilon.$$

This is quite a lot to take in, so let us break it down into pieces.

Definition 1.7.2 (The typical 3 pieces of a definition).

Usually a definition can be broken down into three pieces.

- Scene setting — define symbols and any restrictions on the objects that we are talking about.
- Naming — state the name and any notation for the property or object that the definition is about.
- Properties and restrictions — this is the heart of the definition where we explain to the reader what it is that the object (in our case a function) has to do in order to satisfy the definition.

Let us go back to the definition and look at each of these pieces in turn.

- Setting things up — The first sentence of the definition is really just setting up the picture. It is telling us what the definition is about and sorting out a few technical details.
 - **Let $a \in \mathbb{R}$** — This simply tells us that the symbol “ a ” is a real number³³.
 - **Let $f(x)$ be a function** — This is just setting the scene so that we understand all of the terms and symbols.
 - **defined everywhere in a neighbourhood of a , except possibly at a** — This is just a technical requirement; we need our function to be defined in a little region³⁴ around a . The function doesn’t have to be defined everywhere, but it must be defined for all x -values a little less than a and a little more than a . The definition does not care about what the function does outside this little window, nor does it care what happens exactly at a .
- Names, phrases and notation — The next part of the definition is simply naming the property we are discussing and tells us how to write it down. i.e. we are talking about “limits” and we write them down using the symbols indicated.
- The heart of things — we explain this at length below, but for now we will give a quick explanation. **Work on these two points. They are hard.**
 - **for all $\epsilon > 0$ there exists $\delta > 0$** — It is important we read this in order. It means that we can pick any positive number ϵ we want and there will always be another positive number δ that is going to make what ever follows be true.

33 The symbol “ $\in$ ” is read as “is an element of” — it is definitely not the same as e or ϵ or ε . If you do not recognise “ $\mathbb{R}$ ” or understand the difference between $\mathbb{R}$ and R , then please go back and read Chapter 0⁰ carefully.

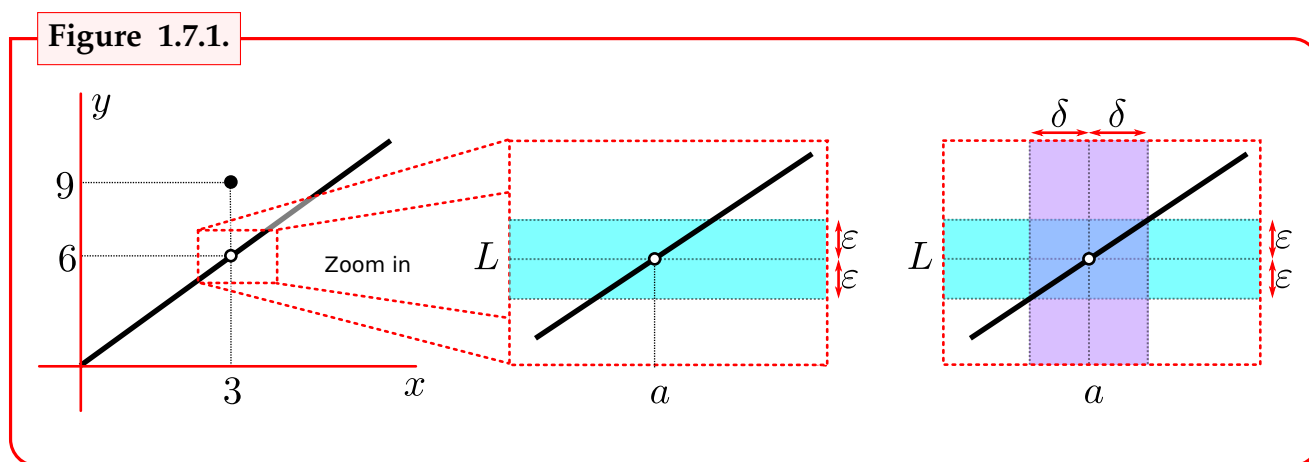
34 The term “neighbourhood of a ” means a small open interval around a — for example $(a - 0.01, a + 0.01)$. Typically we don’t really care how big this little interval is.

- if $0 < |x - a| < \delta$ **then** $|f(x) - L| < \epsilon$ — From the previous point we have our two numbers — any $\epsilon > 0$ then based on that choice of ϵ we have a positive number δ . The current statement says that whenever we have chosen x so that it is very close to a , then $f(x)$ has to be very close to L . How close is “very close”? Well $0 < |x - a| < \delta$ means that x has to be within a distance δ of a (but not exactly a) and similarly $|f(x) - L| < \epsilon$ means that $f(x)$ has to be within a distance ϵ of L .

That is the definition broken up into pieces which hopefully now make more sense, but what does it actually *mean*? Consider a function we saw earlier

$$f(x) = \begin{cases} 2x & x \neq 3 \\ 9 & x = 3 \end{cases}$$

and sketch it again:



We know (from our earlier work) that $\lim_{x \rightarrow 3} f(x) = 6$, so zoom in around $(x, y) = (3, 6)$. To make this look more like our definition, we have $a = 3$ and $L = 6$.

- Pick some small number $\epsilon > 0$ and highlight the horizontal strip of all points (x, y) for which $|y - L| < \epsilon$. This means all the y -values have to satisfy $L - \epsilon < y < L + \epsilon$.
- You can see that the graph of the function passes through this strip for some x -values close to a . What we need to be able to do is to pick a vertical strip of x -values around a so that the function lies inside the horizontal strip.
- That is, we must find a small number $\delta > 0$ so that for any x -value inside the vertical strip $a - \delta < x < a + \delta$, *except exactly at $x = a$* , the value of the function lies inside the horizontal strip, namely $L - \epsilon < y = f(x) < L + \epsilon$.
- We see (pictorially) that we can do this. If we were to choose a smaller value of ϵ making the horizontal strip narrower, it is clear that we can choose the vertical strip to be narrower. Indeed, it doesn't matter how small we make the horizontal strip, we will always be able to construct the second vertical strip.

The above is a pictorial argument, but we can quite easily make it into a mathematical one. We want to show the limit is 6. That means for any ϵ we need to find a δ so that when

$$3 - \delta < x < 3 + \delta \text{ with } x \neq 3 \quad \text{we have} \quad 6 - \epsilon < f(x) < 6 + \epsilon$$

Now we note that when $x \neq 3$, we have $f(x) = 2x$ and so

$$6 - \epsilon < f(x) < 6 + \epsilon \quad \text{implies that} \quad 6 - \epsilon < 2x < 6 + \epsilon$$

this nearly specifies a range of x values, we just need to divide by 2

$$3 - \epsilon/2 < x < 3 + \epsilon/2$$

Hence if we choose $\delta = \epsilon/2$ then we get the desired inequality

$$3 - \delta < x < 3 + \delta$$

i.e. — no matter what $\epsilon > 0$ is chosen, if we put $\delta = \epsilon/2$ then when $3 - \delta < x < 3 + \delta$ with $x \neq 3$ we will have $6 - \epsilon < f(x) < 6 + \epsilon$. This is exactly what we need to satisfy the definition of “limit” above.

The above work gives us the argument we need, but it still needs to be written up properly. We do this below.

Example 1.7.3

Find the limit as $x \rightarrow 3$ of the following function

$$f(x) = \begin{cases} 2x & x \neq 3 \\ 9 & x = 3 \end{cases}$$

Proof. We will show that the limit is equal to 6. Let $\epsilon > 0$ and $\delta = \epsilon/2$. It remains to show that $|f(x) - 6| < \epsilon$ whenever $|x - 3| < \delta$.

So assume that $|x - 3| < \delta$, and so

$$\begin{aligned} 3 - \delta < x < 3 + \delta & \quad \text{multiply both sides by 2} \\ 6 - 2\delta < 2x < 6 + 2\delta \end{aligned}$$

Recall that $f(x) = 2x$ and that since $\delta = \epsilon/2$

$$6 - \epsilon < f(x) < 6 + \epsilon.$$

We can conclude that $|f(x) - 6| < \epsilon$ as required. □

Example 1.7.3

Because of the ϵ and δ in the definition of limits, we need to have ϵ and δ in the proof. While ϵ and δ are just symbols playing particular roles, and could be replaced with other symbols, this style of proof is usually called ϵ - δ proof.

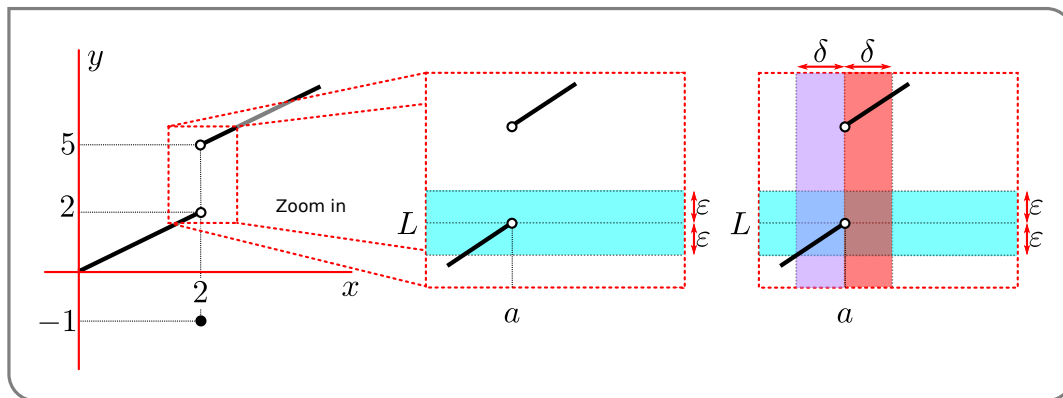
In the above example everything works, but it can be very instructive to see what happens in an example that doesn't work.

Example 1.7.4

Look again at the function

$$f(x) = \begin{cases} x & x < 2 \\ -1 & x = 2 \\ x + 3 & x > 2 \end{cases}$$

and let us see why, according to the definition of the limit, that $\lim_{x \rightarrow 2} f(x) \neq 2$. Again, start by sketching a picture and zooming in around $(x, y) = (2, 2)$:



Try to proceed through the same steps as before:

- Pick some small number $\epsilon > 0$ and highlight a horizontal strip that contains all y -values with $|y - L| < \epsilon$. This means all the y -values have to satisfy $L - \epsilon < y < L + \epsilon$.
- You can see that the graph of the function passes through this strip for some x -values close to a . To the left of a , we can always find some x -values that make the function sit inside the horizontal- ϵ -strip. However, unlike the previous example, there is a problem to the right of a . Even for x -values just a little larger than a , the value of $f(x)$ lies well outside the horizontal- ϵ -strip.
- So given this choice of ϵ , we can find a $\delta > 0$ so that for x inside the vertical strip $a - \delta < x < a$, the value of the function sits inside the horizontal- ϵ -strip.
- Unfortunately, there is no way to choose a $\delta > 0$ so that for x inside the vertical strip $a < x < a + \delta$ (with $x \neq a$) the value of the function sits inside the horizontal- ϵ -strip.
- So it is impossible to choose δ so that for x inside the vertical strip $a - \delta < x < a + \delta$ the value of the function sits inside the horizontal strip $L - \epsilon < y = f(x) < L + \epsilon$.
- Thus the limit of $f(x)$ as $x \rightarrow 2$ is not 2.

Example 1.7.4

Doing things formally with ϵ 's and δ 's is quite painful for general functions. It is far better to make use of the arithmetic of limits (Theorem [1.4.2](#)) and some basic building

blocks (like those in Theorem 1.4.1). Thankfully for most of the problems we deal with in calculus (at this level at least) can be approached in exactly this way.

This does leave the problem of proving the arithmetic of limits and the limits of the basic building blocks. The proof of the Theorem 1.4.2 is quite involved and we leave it to the very end of this Chapter. Before we do that we will prove Theorem 1.4.1 by a formal ϵ - δ proof. Then in the next section we will look at the formal definition of limits at infinity and prove Theorem 1.5.3. The proof of the Theorem 1.5.9, the arithmetic of infinite limits, is very similar to that of Theorem 1.4.2 and so we do not give it.

So let us now prove Theorem 1.4.1 in which we stated two simple limits:

$$\lim_{x \rightarrow a} c = c \qquad \text{and} \qquad \lim_{x \rightarrow a} x = a.$$

Here is the formal ϵ - δ proof:

Proof of Theorem 1.4.1 Since there are two limits to prove, we do each in turn. Let a, c be real numbers.

- Let $\epsilon > 0$ and set $f(x) = c$. Choose $\delta = 1$, then for any x satisfying $|x - a| < \delta$ (or indeed any real number x at all) we have $|f(x) - c| = 0 < \epsilon$. Hence $\lim_{x \rightarrow a} c = c$ as required.
- Let $\epsilon > 0$ and set $f(x) = x$. Choose $\delta = \epsilon$, then for any x satisfying $|x - a| < \delta$ we have

$$\begin{aligned} a - \delta < x < a + \delta \text{ but } f(x) = x \text{ and } \delta = \epsilon \text{ so} \\ a - \epsilon < f(x) < a + \epsilon \end{aligned}$$

Thus we have $|f(x) - a| < \epsilon$. Hence $\lim_{x \rightarrow a} x = a$ as required.

This completes the proof. □

1.8 ▲ (Optional) — Making Infinite Limits a Little More Formal

For those of you who made it through the formal $\epsilon - \delta$ definition of limits we give the formal definition of limits involving infinity:

Definition 1.8.1 (Limits involving infinity — formal).

(a) Let f be a function defined on the whole real line. We say that

the limit as x approaches ∞ of $f(x)$ is L

or equivalently

$f(x)$ converges to L as x goes to ∞

and write

$$\lim_{x \rightarrow \infty} f(x) = L$$

if and only if for every $\epsilon > 0$ there exists $M \in \mathbb{R}$ so that $|f(x) - L| < \epsilon$ whenever $x > M$.

Similarly we write

$$\lim_{x \rightarrow -\infty} f(x) = K$$

if and only if for every $\epsilon > 0$ there exists $N \in \mathbb{R}$ so that $|f(x) - K| < \epsilon$ whenever $x < N$.

(b) Let a be a real number and $f(x)$ be a function defined for all $x \neq a$. We write

$$\lim_{x \rightarrow a} f(x) = \infty$$

if and only if for every $P > 0$ there exists $\delta > 0$ so that $f(x) > P$ whenever $0 < |x - a| < \delta$.

(c) Let f be a function defined on the whole real line. We write

$$\lim_{x \rightarrow \infty} f(x) = \infty$$

if and only if for every $P > 0$ there exists $M > 0$ so that $f(x) > P$ whenever $x > M$.

Note that we can loosen the above requirements on the domain of definition of f — for example, in part (a) all we actually require is that $f(x)$ be defined for all x larger than some value. It would be sufficient to require “there is some $x_0 \in \mathbb{R}$ so that $f(x)$ is defined for all $x > x_0$ ”. Also note that there are obvious variations of parts (b) and (c) with ∞ replaced by $-\infty$.

For completeness let’s prove Theorem 1.5.3 using this formal definition. The layout of the proof will be very similar to our proof of Theorem 1.4.1.

Proof of Theorem 1.5.3 There are four limits to prove in total and we do each in turn. Let $c \in \mathbb{R}$.

- Let $\epsilon > 0$ and set $f(x) = c$. Choose $M = 0$, then for any x satisfying $x > M$ (or indeed any real number x at all) we have $|f(x) - c| = 0 < \epsilon$. Hence $\lim_{x \rightarrow \infty} c = c$ as required.
- The proof that $\lim_{x \rightarrow -\infty} c = c$ is nearly identical. Again, let $\epsilon > 0$ and set $f(x) = c$. Choose $N = 0$, then for any x satisfying $x < N$ we have $|f(x) - c| = 0 < \epsilon$. Hence $\lim_{x \rightarrow -\infty} c = c$ as required.
- Let $\epsilon > 0$ and set $f(x) = x$. Choose $M = \frac{1}{\epsilon}$. Then when $x > M$ we have

$$\begin{aligned} 0 < M < x & \qquad \text{divide through by } xM \text{ to get} \\ 0 < \frac{1}{x} < \frac{1}{M} = \epsilon \end{aligned}$$

Since $x > 0$, $1/x = |1/x| = |1/x - 0| < \epsilon$ as required.

- Again, the proof in the limit to $-\infty$ is similar but we have to be careful of signs. Let $\epsilon > 0$ and set $f(x) = x$. Choose $N = -\frac{1}{\epsilon}$. Then when $x < N$ we have

$$\begin{aligned} 0 > N > x & \qquad \text{divide through by } xN \text{ to get} \\ 0 > \frac{1}{x} > \frac{1}{N} = -\epsilon \end{aligned}$$

Notice that by assumption both $x, N < 0$, so $xN > 0$. Now since $x < 0$, $1/x = -|1/x| = |1/x - 0| < \epsilon$ as required.

This completes the proof. □

1.9 ▲ (Optional) — Proving the Arithmetic of Limits

Perhaps the most useful theorem of this chapter is Theorem [1.4.2](#) which shows how limits interact with arithmetic. In this (optional) section we will prove both the arithmetic of limits Theorem [1.4.2](#) and the Squeeze Theorem [1.4.17](#). Before we get to the proofs it is very helpful to prove three technical lemmas that we'll need. The first is a very general result about absolute values of numbers:

Lemma 1.9.1 (The triangle inequality).

For any $x, y \in \mathbb{R}$

$$|x + y| \leq |x| + |y|$$

Proof. Notice that for any real number x , we always have $-x, x \leq |x|$ and either $|x| = x$ or $|x| = -x$. So now let $x, y \in \mathbb{R}$. Then we must have either

$$|x + y| = x + y \leq |x| + |y|$$

or

$$|x + y| = -x - y \leq |x| + |y|$$

In both cases we end up with $|x + y| \leq |x| + |y|$. $\square$

The second lemma is more specialised. It proves that if we have a function $f(x) \rightarrow F$ as $x \rightarrow a$ then there must be a small window around $x = a$ where the function $f(x)$ must only take values not far from F . In particular it tells us that $|f(x)|$ cannot be bigger than $|F| + 1$ when x is very close to a .

Lemma 1.9.2.

Let $a \in \mathbb{R}$ and let f be a function so that $\lim_{x \rightarrow a} f(x) = F$. Then there exists a $\delta > 0$ so that if $0 < |x - a| < \delta$ then we also have $|f(x)| \leq |F| + 1$.

The proof is mostly just manipulating the ϵ - δ definition of a limit with $\epsilon = 1$.

Proof. Let $\epsilon = 1$. Then since $f(x) \rightarrow F$ as $x \rightarrow a$, there exists a $\delta > 0$ so that when $0 < |x - a| < \delta$, we also have $|f(x) - F| \leq \epsilon = 1$. So now assume $0 < |x - a| < \delta$. Then

$$\begin{aligned} -\epsilon &\leq f(x) - F \leq \epsilon && \text{rearrange a little} \\ -\epsilon + F &\leq f(x) \leq \epsilon + F \end{aligned}$$

Now $\epsilon + F \leq \epsilon + |F|$ and $-\epsilon + F \geq -\epsilon - |F|$, so

$$-\epsilon - |F| \leq f(x) \leq \epsilon + |F|$$

Hence we have $|f(x)| \leq \epsilon + |F| = |F| + 1$. $\square$

Finally our third technical lemma gives us a bound in the other direction; it tells us that when x is close to a , the value of $|f(x)|$ cannot be much smaller than $|F|$.

Lemma 1.9.3.

Let $a \in \mathbb{R}$ and $F \neq 0$ and let f be a function so that $\lim_{x \rightarrow a} f(x) = F$. Then there exists $\delta > 0$ so that when $0 < |x - a| < \delta$, we have $|f(x)| > |F|/2$.

Proof. Set $\epsilon = |F|/2 > 0$. Then since $f(x) \rightarrow F$, we know there exists a $\delta > 0$ so that when $0 < |x - a| < \delta$ we have $|f(x) - F| < \epsilon$. So now assume $0 < |x - a| < \delta$ so that $|f(x) - F| < \epsilon = |F|/2$. Then

$$\begin{aligned} |F| &= |F - f(x) + f(x)| && \text{sneaky trick} \\ &\leq |f(x) - F| + |f(x)| && \text{but } |f(x) - F| < \epsilon \\ &< \epsilon + |f(x)| \end{aligned}$$

Hence $|f(x)| > |F| - \epsilon = |F|/2$ as required. $\square$

Now we are in a position to prove Theorem [1.4.2](#). The proof has more steps than the previous $\epsilon - \delta$ proofs we have seen. This is mostly because we do not have specific functions $f(x)$ and $g(x)$ and instead must play with them in the abstract — and make good use of the formal definition of limits.

We will break the proof into three pieces. The minimum that is required is to prove that

$$\begin{aligned}\lim_{x \rightarrow a} (f(x) + g(x)) &= F + G \\ \lim_{x \rightarrow a} f(x) \cdot g(x) &= F \cdot G \\ \lim_{x \rightarrow a} 1/g(x) &= 1/G \quad \text{if } G \neq 0.\end{aligned}$$

From these three we can prove that

$$\begin{aligned}\lim_{x \rightarrow a} f(x) \cdot c &= F \cdot c \\ \lim_{x \rightarrow a} (f(x) - g(x)) &= F - G \\ \lim_{x \rightarrow a} f(x)/g(x) &= F/G \quad \text{if } G \neq 0.\end{aligned}$$

The first follows by setting $g(x) = c$ and using $\lim f(x) \cdot g(x)$. The second follows by setting $c = -1$, putting $h(x) = (-1) \cdot g(x)$ and then applying both $\lim f(x) \cdot g(x)$ and $\lim f(x) + g(x)$. The third follows by setting $h(x) = 1/g(x)$ and then using $\lim f(x) \cdot h(x)$.

Starting with addition, in order to satisfy the definition of limit, we are going to have to show that

$$|(f(x) + g(x)) - (F + G)| \text{ is small}$$

when we know that $|f(x) - F|, |g(x) - G|$ are small. To do this we use the triangle inequality above showing that

$$|(f(x) + g(x)) - (F + G)| = |(f(x) - F) + (g(x) - G)| \leq |f(x) - F| + |g(x) - G|$$

This is the key technical piece of the proof. So if we want the LHS of the above to be size ϵ , we need to make sure that each term on the RHS is of size $\epsilon/2$. The rest of the proof is setting up facts based on the definition of limits and then rearranging facts to reach the conclusion.

Proof. Proof of Theorem [1.4.2](#) — limit of a sum. Let $a \in \mathbb{R}$ and assume that

$$\lim_{x \rightarrow a} f(x) = F \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = G.$$

We wish to show that

$$\lim_{x \rightarrow a} f(x) + g(x) = F + G.$$

Let $\epsilon > 0$ — we have to find a $\delta > 0$ so that when $|x - a| < \delta$ we have $|(f(x) + g(x)) - (F + G)| < \epsilon$.

Let $\epsilon > 0$ and set $\epsilon_1 = \epsilon_2 = \epsilon/2$. By the definition of limits, because $f(x) \rightarrow F$ there exists some $\delta_1 > 0$ so that whenever $|x - a| < \delta_1$, we also have $|f(x) - F| < \epsilon_1$. Similarly

there exists $\delta_2 > 0$ so that if $|x - a| < \delta_2$, then we must have $|g(x) - G| < \epsilon_2$. So now choose $\delta = \min\{\delta_1, \delta_2\}$ and assume $|x - a| < \delta$. Then we must have that $|x - a| < \delta_1, \delta_2$ and so we also have

$$|f(x) - F| < \epsilon_1 \qquad |g(x) - G| < \epsilon_2$$

Now consider $|(f(x) + g(x)) - (F + G)|$ and rearrange the terms:

$$\begin{aligned} |(f(x) + g(x)) - (F + G)| &= |(f(x) - F) + (g(x) - G)| \quad \text{now apply triangle inequality} \\ &\leq |f(x) - F| + |g(x) - G| \quad \text{use facts from above} \\ &< \epsilon_1 + \epsilon_2 \\ &= \epsilon. \end{aligned}$$

Hence we have shown that for any $\epsilon > 0$ there exists some $\delta > 0$ so that when $|x - a| < \delta$ we also have $|(f(x) + g(x)) - (F + G)| < \epsilon$. Which is exactly the formal definition of the limit we needed to prove. $\square$

Let us do similarly for the limit of a product. Some of the details of the proof are very similar, but there is a little technical trick in the middle to make it work. In particular we need to show that

$$|f(x) \cdot g(x) - F \cdot G| \text{ is small}$$

when we know that $|f(x) - F|$ and $|g(x) - G|$ are both small. Notice that

$$\begin{aligned} f(x) \cdot g(x) - F \cdot G &= f(x) \cdot g(x) - F \cdot G + \underbrace{f(x) \cdot G - f(x) \cdot G}_{=0} \\ &= f(x) \cdot g(x) - f(x) \cdot G + f(x) \cdot G - F \cdot G \\ &= f(x) \cdot (g(x) - G) + (f(x) - F) \cdot G \end{aligned}$$

So if we know $|f(x) - F|$ is small and $|g(x) - G|$ is small then we are done — except that we also need to know that $f(x)$ doesn't become really large near a — this is exactly why we needed to prove Lemma [1.9.2](#).

As was the case in the previous proof, we want the LHS to be of size at most ϵ , so we want, for example, the two terms on the RHS to be of size at most $\epsilon/2$. This means

- we need $|G| \cdot |f(x) - F|$ to be of size at most $\epsilon/2$, and
- we need $|g(x) - G|$ to be of size at most $\epsilon/2(|F|+1)$ since we know that $|f(x)| \leq |F| + 1$ when x is close to a .

Armed with these tricks we turn to the proofs.

Proof. Proof of Theorem [1.4.2](#) — limit of a product. Let $a \in \mathbb{R}$ and assume that

$$\lim_{x \rightarrow a} f(x) = F \qquad \text{and} \qquad \lim_{x \rightarrow a} g(x) = G.$$

We wish to show that

$$\lim_{x \rightarrow a} f(x) \cdot g(x) = F \cdot G.$$

Let $\epsilon > 0$. Set $\epsilon_1 = \frac{\epsilon}{2(|G|+1)}$ (the extra $+1$ in the denominator is just there to make sure that ϵ_1 is well-defined even if $G = 0$), and $\epsilon_2 = \frac{\epsilon}{2(|F|+1)}$. From this we establish the existence of $\delta_1, \delta_2, \delta_3$ which we need below.

- By assumption $f(x) \rightarrow F$ so there exists $\delta_1 > 0$ so that whenever $|x - a| < \delta_1$, we also have $|f(x) - F| < \epsilon_1$.
- Similarly because $g(x) \rightarrow G$, there exists $\delta_2 > 0$ so that whenever $|x - a| < \delta_2$, we also have $|g(x) - G| < \epsilon_2$.
- By Lemma 1.9.2 there exists $\delta_3 > 0$ so that whenever $|x - a| < \delta_3$, we also have $|f(x)| \leq |F| + 1$.

Let $\delta = \min\{\delta_1, \delta_2, \delta_3\}$, assume $|x - a| < \delta$ and consider $|f(x) \cdot g(x) - F \cdot G|$. Rearrange the terms as we did above:

$$\begin{aligned} |f(x) \cdot g(x) - F \cdot G| &= |f(x) \cdot (g(x) - G) + (f(x) - F) \cdot G| \\ &\leq |f(x)| \cdot |g(x) - G| + |G| \cdot |f(x) - F| \end{aligned}$$

By our three dot-points above we know that $|f(x) - F| < \epsilon_1$ and $|g(x) - G| < \epsilon_2$ and $|f(x)| \leq |F| + 1$, so we have

$$\begin{aligned} |f(x) \cdot g(x) - F \cdot G| &< |f(x)| \cdot \epsilon_2 + |G| \cdot \epsilon_1 && \text{sub in } \epsilon_1, \epsilon_2 \text{ and bound on } f(x) \\ &< (|F| + 1) \cdot \frac{\epsilon}{2(|F| + 1)} + |G| \cdot \frac{\epsilon}{2(|G| + 1)} \\ &\leq \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon. \end{aligned}$$

Thus we have shown that for any $\epsilon > 0$ there exists $\delta > 0$ so that when $|x - a| < \delta$ we also have $|f(x) \cdot g(x) - F \cdot G| < \epsilon$. Hence $f(x) \cdot g(x) \rightarrow F \cdot G$. $\square$

Finally we can prove the limit of a reciprocal. Notice that

$$\frac{1}{g(x)} - \frac{1}{G} = \frac{G - g(x)}{g(x) \cdot G}$$

We need to show the LHS is of size at most ϵ when x is close enough to a , so if $G - g(x)$ is small we are done — except if $g(x)$ or G are close to zero. By assumption (go back and read Theorem 1.4.2) we have $G \neq 0$, and we know from Lemma 1.9.3 that $|g(x)|$ cannot be smaller than $|G|/2$. Together these imply that the denominator on the RHS cannot be zero and indeed must be of magnitude at least $|G|^2/2$. Thus we need $|G - g(x)|$ to be of size at most $\epsilon \cdot |G|^2/2$.

Proof. Proof of Theorem 1.4.2 — limit of a reciprocal. Let $\epsilon > 0$ and set $\epsilon_1 = \epsilon|G|^2 \cdot \frac{1}{2}$. We now use this and Lemma 1.9.3 to establish the existence of δ_1, δ_2 .

- Since $g(x) \rightarrow G$ we know that there exists $\delta_1 > 0$ so that when $|x - a| < \delta_1$ we also have $|g(x) - G| < \epsilon_1$.
- By Lemma 1.9.3 there exists δ_2 so that when $|x - a| < \delta_2$ we also have $|g(x)| > |G|/2$. Equivalently, when $|x - a| < \delta_2$ we also have $\left| \frac{G}{2g(x)} \right| < 1$.

Set $\delta = \min\{\delta_1, \delta_2\}$ and assume $|x - a| < \delta$. Then

$$\begin{aligned}
 \left| \frac{1}{g(x)} - \frac{1}{G} \right| &= \left| \frac{G - g(x)}{g(x) \cdot G} \right| \\
 &= |g(x) - G| \cdot \frac{1}{|G| \cdot |g(x)|} && \text{by assumption} \\
 &< \frac{\epsilon_1}{|G| \cdot |g(x)|} && \text{sub in } \epsilon_1 \\
 &= \epsilon \cdot \frac{|G|}{2|g(x)|} && \text{since } \left| \frac{G}{2g(x)} \right| < 1 \\
 &< \epsilon
 \end{aligned}$$

Thus we have shown that for any $\epsilon > 0$ there exists $\delta > 0$ so that when $|x - a| < \delta$ we also have $|1/g(x) - 1/G| < \epsilon$. Hence $1/g(x) \rightarrow 1/G$. $\square$

Proof. Proof of Theorem [1.4.17](#) — Squeeze / sandwich / pinch. In the squeeze theorem, we are given three functions $f(x)$, $g(x)$ and $h(x)$ and are told that

$$f(x) \leq g(x) \leq h(x) \quad \text{and} \quad \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L$$

and we must conclude from this that $\lim_{x \rightarrow a} g(x) = L$ too. That is, we are given some fixed, but unspecified, $\epsilon > 0$ and it is up to us to find a $\delta > 0$ with the property that $|g(x) - L| < \epsilon$ whenever $|x - a| < \delta$. Now because we have been told that f and h both converge to L , there exist $\delta_1 > 0$ and $\delta_2 > 0$ such that

- $|f(x) - L| < \epsilon$, i.e. $L - \epsilon < f(x) < L + \epsilon$, whenever $|x - a| < \delta_1$, and
- $|h(x) - L| < \epsilon$, i.e. $L - \epsilon < h(x) < L + \epsilon$, whenever $|x - a| < \delta_2$

So set $\delta = \min\{\delta_1, \delta_2\}$ and assume $|x - a| < \delta$. Then both $L - \epsilon < f(x) < L + \epsilon$ and $L - \epsilon < h(x) < L + \epsilon$ so that

$$\begin{aligned}
 L - \epsilon &< f(x) \leq g(x) \leq h(x) < L + \epsilon && \text{which implies that} \\
 L - \epsilon &< g(x) < L + \epsilon && \text{which in turn gives us} \\
 |g(x) - L| &< \epsilon
 \end{aligned}$$

as desired. $\square$